25901194 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECSC202 - SYSTEM PROGRAMMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Identify the role of the symbol table in Pass-1. [K] / [CO1]
2. Provide an example assembly instruction with all fields labeled [K] / [CO1]
3. Define a macro. [K] / [CO2]
4. Interpret the compilation of control structures. [U] / [CO2]
5. Define relocation table. [K] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **5** marks

6. Compare one-pass and two-pass assembly schemes. [U] / [CO1]
7. Explain the parsing process using a simple grammar and diagram. [U] / [CO1]

8. List the functions of system software and explain any three. [K] / [CO1]

9. Illustrate the use of debugging tools in detecting logical and runtime errors. [A] / [CO2]

10. Differentiate between interpreters and compilers based on execution and memory usage. [U] / [CO2]

11. Describe the working of a relocating loader. [U] / [CO3]

12. Explain the role of overlay manager. [U] / [CO3]

[5x5 = 25]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

13. Describe the design and working of a macro preprocessor with neat diagrams. [U] / [CO2]

14. Analyze the concepts of linking and relocation with suitable examples and diagrams. [An] / [CO3]

[15x1 = 15]